BIO 4AE3. 24 NOV 2025

EMERGING. new.

ZOONOTIC. spillovers (?)
($R_0 < 1$)

could also refer.
expanding its range
becoming more infectious
more virulent
antibiotic resistant

RE-EMERGING DISEASE.
previously controlled but
is resurging
(socioeconomic change,
disruption of society, ...)
biology, landscape change,
climate change, socioeconomic change

ENCOUNTER + COMPATIBILITY filters
(landscape etc.) (biological)

- DISEASE TRIANGLE.

host, parasite, environment
(plant pathology).

LOCAL > GLOBAL

global change can ~~make~~ be good or bad for disease

- MALARIA in NORTH AMERICA.
  - brought to the 'New World' by European colonists
  - especially by west African enslaved people.

well established the 1850s.
poorly diagnosed, poorly ~~treated~~ treated.
DECLINED 1900 - 1920 (drainage)
1930s. 5000 deaths/year (US)
1950s. < 10 deaths/year
Improved NUTRITION, medical care, housing, treatment.

R_0 for vector-borne diseases: PROPORTIONAL to vector pop. density

MALARIA + local landscape change:
- western Kenya:
  malaria prevalence increased
  ⇒ ? due to availability of brick-making pits

- VECTOR POP DYNAMICS:
  temperature
  hydrology
  - predators.

- DEFORESTATION in Amazon.
  increase malaria prevalence
  (due to forest edge habitat)
  increased human-vector contact
  ↳ MALARIA decreases deforestation.

- mosquito-borne disease in temperate countries.
- expansion of vector ranges?

spillover (humans get infected from animals) → Local transmission events
autochthonous.
$R_0 > 0$

MALARIA. Anopheles mosquitoes.
chikungunya, Zika, dengue, yellow fever, ... Aedes.
Aedes aegypti. A. albopictus.

climate change on malaria range?
- Indian tea plantations.
  - what's the actual change in temp?

temp changes:
Anopheles → Aedes.
malaria → dengue, yellow fever, chikungunya.